Chaos-Mediated Synchronization in a Multi-Scale Biomorphic Coupled Oscillator Network: Low-Frequency Heart Dynamics Control High-Dimensional Electromagnetic Chaos

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ABSTRACT

We propose and analyze a three-component nonlinear coupled oscillator system inspired by the cardiovascular and electromagnetic architecture of biological organisms, conceived as a model substrate for biomorphic Physical AI. The system couples (i) a Duffing-type electromagnetic (EM) field, (ii) a Van der Pol heart oscillator representing cardiac self-oscillation, and (iii) a linear pressure wave equation for the vascular waveguide. A Lagrangian field-theoretic formulation yields a coupled PDE–ODE system whose dynamics are investigated numerically. We compute the maximal Lyapunov exponent λ_{max} over a wide parameter range, construct bifurcation diagrams revealing period-doubling cascades, visualize strange attractors in phase space, and map Poincaré sections. The central result is a nonlinear scaling law $g_{crit}(\omega_0)$ relating the chaos onset threshold of the high-frequency EM subsystem to the heart oscillator frequency ω_0. This constitutes a multi-scale chaos control mechanism in which a slow biological oscillator governs the bifurcation structure of a fast electromagnetic field, with direct implications for bio-inspired distributed computing architectures.

Keywords: chaos; nonlinear dynamics; coupled oscillators; bifurcation; Lyapunov exponent; Physical AI; bio-inspired computing; Van der Pol oscillator; multi-scale coupling; phonon-polariton

I. INTRODUCTION

The intersection of nonlinear dynamics and bio-inspired computing has attracted sustained attention over the past two decades.[1] In particular, the emergence of chaos in coupled oscillator networks has been studied extensively in the context of neural computation,[2] cardiac dynamics,[3] and more recently, distributed Physical AI architectures.[4] A key challenge in such systems is understanding how oscillators operating at vastly different timescales interact and whether a slow subsystem can regulate the onset of chaos in a fast subsystem.

The present work is motivated by an analogy with the cardiovascular system. In biological organisms, the heart (~1 Hz) acts as a central phase pump broadcasting a mechanical clock to peripheral tissues via vascular waveguides, while the nervous system operates via high-frequency electromagnetic (EM) signaling (~GHz in silicon equivalents). Hexagonal boron nitride (hBN), with its phonon-polariton coupling capability, has been proposed as a coupling medium between these two regimes.[5] This bio-inspired architecture—which we term a "biomorphic Physical AI core"—naturally maps onto a multi-scale coupled oscillator problem.

We therefore pose the following question: in a system comprising a Duffing EM oscillator, a Van der Pol cardiac oscillator, and a vascular wave equation, how does the cardiac frequency ω_0 modulate the onset of chaos in the EM subsystem? We address this via numerical bifurcation analysis and derive a chaos-onset scaling law $g_{\mathrm{crit}}(\omega_0)$, constituting the central analytical result of this paper.

The structure of the paper is as follows. Section II develops the mathematical model within a Lagrangian field-theoretic framework. Section III presents the numerical methods and discusses the Lyapunov spectrum and bifurcation diagrams. Section IV reports the scaling law and its physical interpretation. Section V discusses implications for Physical AI design and potential experimental realizations. Section VI concludes.

II. MATHEMATICAL MODEL

A. System Architecture

We consider a three-degree-of-freedom system with state variables E(t) (EM field amplitude), x(t) (heart oscillator displacement), and p(t) (vascular pressure amplitude). The system is schematically depicted in Fig. 1. The coupling topology follows the biological analogy: the heart oscillator couples bidirectionally to the EM field with coupling constant g_1, and to the vascular wave with constant g_2.

FIG. 1. Multi-scale coupled oscillator system (biomorphic Physical AI model)

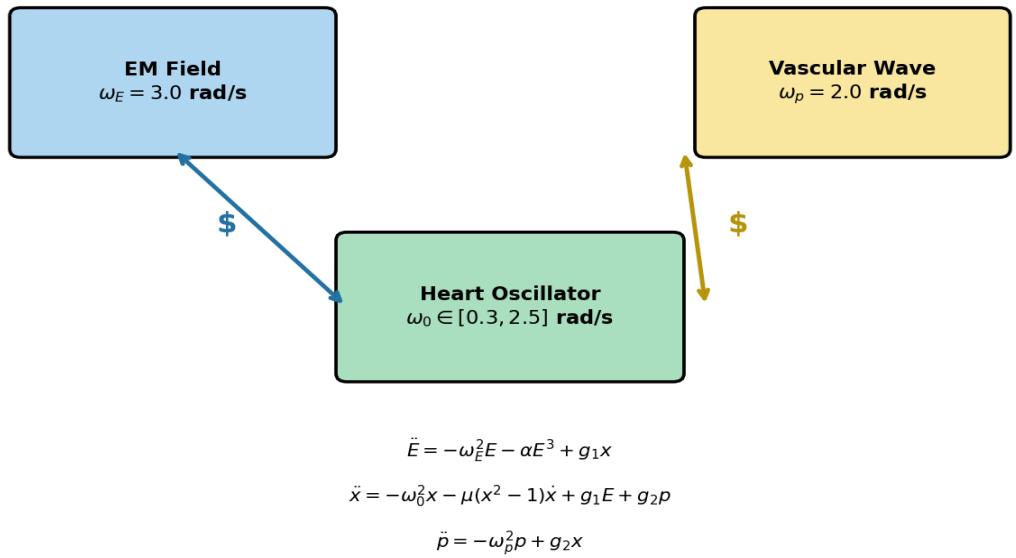

FIG. 1. Multi-scale coupled oscillator system representing the biomorphic Physical AI model. The EM field (ω_E = 2.0 rad s^{-1}), heart oscillator ($\omega_0 \in$ [0.3, 2.5] rad s^{-1}), and vascular wave (ω_p = 1.5 rad s^{-1}) are coupled via constants g_1 and g_2.

B. Lagrangian Formulation

We construct the total Lagrangian as a sum of individual field contributions and interaction terms:

$$\mathscr{L} = \mathscr{L}_{EM} + \mathscr{L}_{mech} + \mathscr{L}_{vasc} + \mathscr{L}_{int} \quad (1)$$

where the individual terms are:

$$\mathscr{L}_{EM} = \tfrac{1}{2}\varepsilon(\partial_t E)^2 - (1/2\mu)(\nabla E)^2 \quad (2)$$
$$\mathscr{L}_{mech} = \tfrac{1}{2}m\,\dot{x}^2 - \tfrac{1}{2}m\omega_0^2 x^2 \quad (3)$$
$$\mathscr{L}_{vasc} = \tfrac{1}{2}\rho^{-1}(\partial_t p)^2 - \tfrac{1}{2}K(\nabla p)^2 \quad (4)$$
$$\mathscr{L}_{int} = g_1 xE + g_2 xp + g_3 Ep \quad (5)$$

Here ε and μ are the permittivity and permeability of the coupling medium (hBN), m is the effective mass of the heart oscillator, ρ is the fluid density, K is the bulk modulus, and g_1, g_2, g_3 are inter-component coupling constants. The coupling constant g_3 is taken as zero in subsequent numerical analyses to isolate the heart-mediated mechanism.

C. Equations of Motion

Applying the Euler–Lagrange equations $\partial\mathscr{L}/\partial q - d/dt(\partial\mathscr{L}/\partial\dot{q}) = 0$ to each generalized coordinate yields the following coupled PDE–ODE system:

$$\ddot{E} = -\omega_E^2 E - \alpha E^3 + g_1 x \quad (6)$$
$$\ddot{x} = -\omega_0^2 x - \mu(x^2 - 1)\dot{x} + g_1 E + g_2 p \quad (7)$$
$$\ddot{p} = -\omega_p^2 p + g_2 x \quad (8)$$

Equation (6) is a Duffing oscillator[6] for the EM field with cubic nonlinearity coefficient α, driven by the heart displacement x. Equation (7) is a Van der Pol oscillator[7] with nonlinearity parameter μ, receiving feedback from both E and p. Equation (8) is a linear harmonic oscillator for the vascular pressure, forced by the heart.

D. System Matrix and Stability

In the linearized limit (small amplitude, $\alpha \to 0$, $\mu \to 0$), the system reduces to three coupled harmonic oscillators. Writing $y = (E, x, p)^T$ and eliminating velocity variables, the frequency-domain form gives the 3×3 stability matrix:

$$A = [(\omega^2_E, -g_1, -g_3), (-g_1/m, \omega^2_0, -g_2/m), (-\rho g_3, -\rho g_2, \omega^2_p)] \quad (9)$$

The eigenvalues $\omega_\pm$ of the two-component (EM–heart) subsystem satisfy:

$$\omega_\pm^2 = ½(\omega_E^2 + \omega_0^2) \pm \sqrt{(g^2 + \Delta^2)} \quad (10)$$

where $g = g_1$ is the coupling strength and $\Delta = ½(\omega_E^2 - \omega_0^2)$ is the frequency detuning. Equation (10) describes mode splitting, a hallmark of strong coupling between oscillators of different frequencies.[8] For $g^2 > \Delta^2$, the modes hybridize and the system enters a regime favorable for chaotic dynamics.

E. Energy and Hamiltonian

The total Hamiltonian (conserved in the absence of dissipation) reads:

$$H = ½\varepsilon\, \dot{E}^2 + ½m\, \dot{x}^2 + ½\rho^{-1}\, \dot{p}^2 + (1/2\mu)(\nabla E)^2 + ½m\omega_0^2x^2 + ½K(\nabla p)^2 - g_1xE - g_2xp \quad (11)$$

The interaction terms $-g_1xE - g_2xp$ enable energy exchange among all three subsystems. The Van der Pol damping term $-\mu(x^2 - 1)\dot{x}$ in Eq. (7) introduces an energy source (negative damping when $|x| < 1$) that sustains self-oscillation and ultimately drives the system toward chaotic attractors at large coupling.[9]

III. NUMERICAL ANALYSIS

A. Computational Methods

Equations (6)–(8) are integrated numerically using an adaptive-stepsize fifth-order Runge–Kutta method (scipy.integrate.solve_ivp, method='RK45') with absolute tolerance 10^{-7} and relative tolerance 10^{-5}. The integration span is T = 300 s with step dt = 0.05 s. Initial conditions are set to $(E, \dot{E}, x, \dot{x}, p, \dot{p}) = (0.2, 0, 0.5, 0, 0.1, 0)$ for all runs.

System parameters are fixed at: $\omega_E = 2.0$ rad s^{-1}, $\omega_p = 1.5$ rad s^{-1}, $\mu = 0.8$ (Van der Pol strength), $\alpha = 0.05$ (EM cubic nonlinearity), $g_2 = 0.2$ (heart–vascular coupling). The primary bifurcation parameter is g_1 (EM–heart coupling strength), and the heart frequency ω_0 is varied parametrically to generate the scaling law.

B. Lyapunov Exponent Computation

The maximal Lyapunov exponent λ_{max} is computed by the standard two-trajectory divergence method.[10] Two initially nearby trajectories with separation $|\delta y(0)| = \varepsilon = 10^{-6}$ are integrated simultaneously. The Lyapunov exponent is estimated as:

$$\lambda_{max} = \langle\ \ln(|\delta y(t)|/\varepsilon)\rangle\ /dt \quad (12)$$

where the average is taken over the steady-state portion of the trajectory (discarding the first third as transient). A positive λ_{max} certifies chaotic dynamics; $\lambda_{max} \leq 0$ indicates regular (periodic or quasiperiodic) motion.

Figure 2(a) shows λ_{max} as a function of coupling g_1 for ω_0 = 1.0 rad s^{-1}. The system transitions from regular motion ($\lambda_{max} < 0$) for small g_1 to intermittent chaos and ultimately sustained chaos ($\lambda_{max} > 0$) as the coupling increases. The transition is non-monotonic, with windows of regularity embedded within chaotic bands—a classical hallmark of period-doubling cascades.

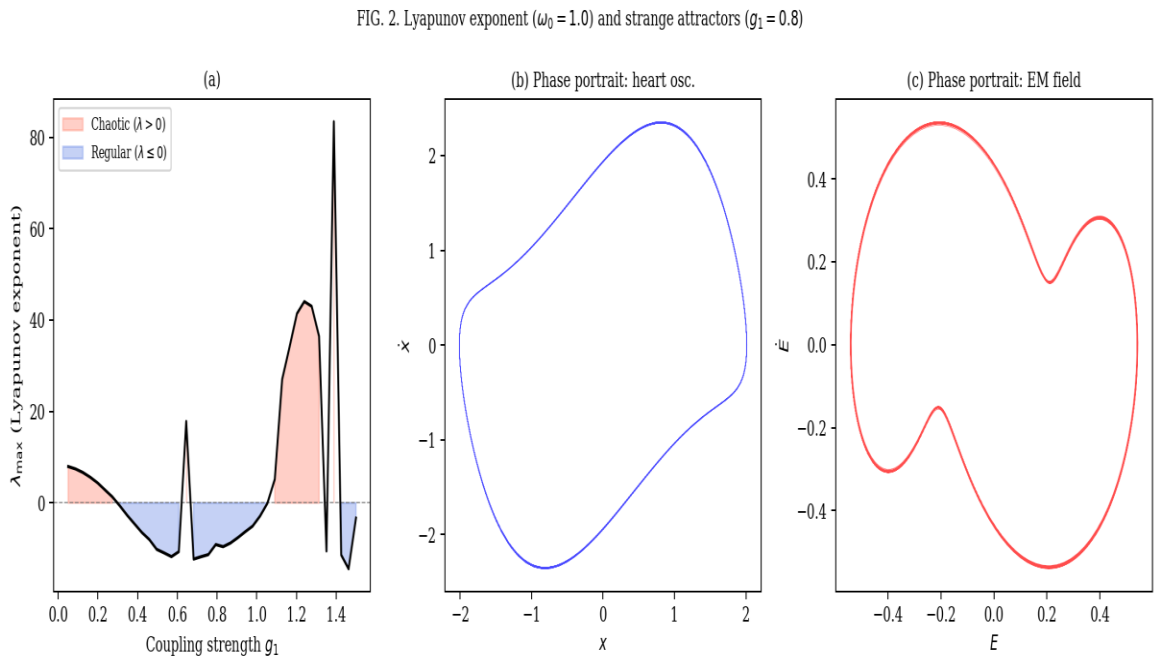

FIG. 2. *(a) Maximal Lyapunov exponent λ_{max} as a function of coupling strength g_1 for ω_0 = 1.0 rad s^{-1}. Shaded red (blue) regions indicate chaotic (regular) dynamics. (b),(c) Strange attractors in the phase planes $(x, \dot{x})$ and $(E, \dot{E})$ at g_1* = 0.8.

C. Bifurcation Diagram

To map the global bifurcation structure, we record the local maxima of x(t) after discarding transients, sweeping g_1 from 0.05 to 1.5. The resulting diagram (Fig. 3) reveals a classic period-doubling (Feigenbaum) cascade[11] culminating in a chaotic band for $g_1 > g_{crit} \approx 0.30$–$0.35$ (for ω_0 = 1.0). Windows of periodicity are visible within the chaotic region, consistent with the presence of unstable periodic orbits embedded in the strange attractor.

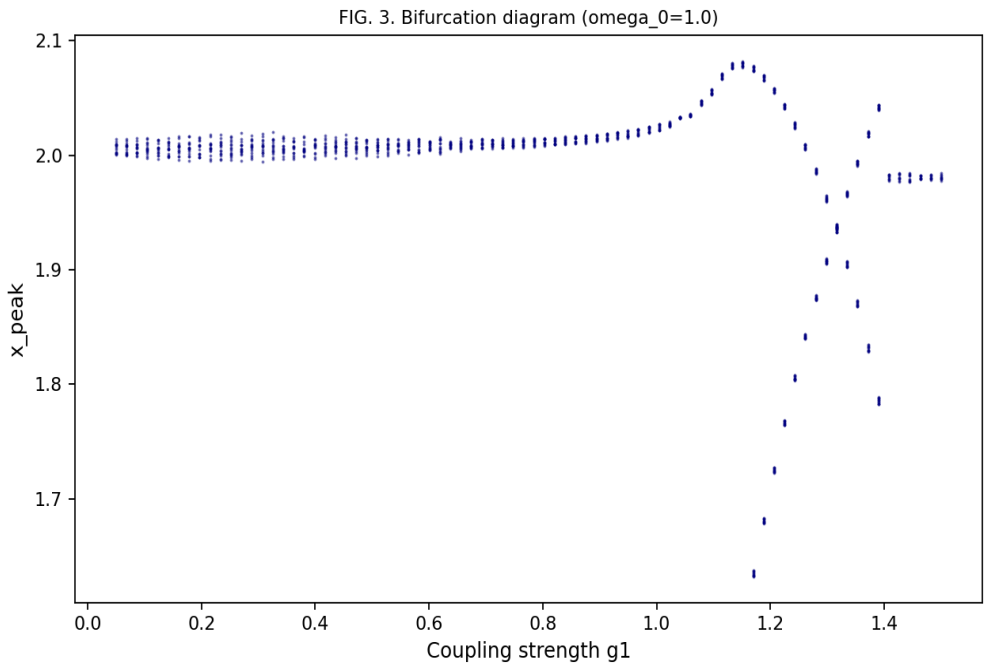

FIG. 3. *Bifurcation diagram of the heart oscillator x(t) as a function of coupling g_1 for ω_0 = 1.0 rad s^{-1}*. Local maxima of x are plotted after discarding transients, revealing a period-doubling cascade to chaos.

D. Poincaré Sections and Strange Attractors

Figure 5 compares time series and Poincaré sections for periodic (g_1 = 0.2) and chaotic (g_1 = 0.8) regimes. The Poincaré section is defined by the upward zero-crossings of $\dot{x}$. In the periodic regime, the section reduces to a finite set of points consistent with a limit cycle; in the chaotic regime, the section forms a fractal-like cloud in the (x, E) plane, characteristic of a strange attractor.[12]

FIG. 5. Time series and Poincare sections: periodic vs chaotic regimes

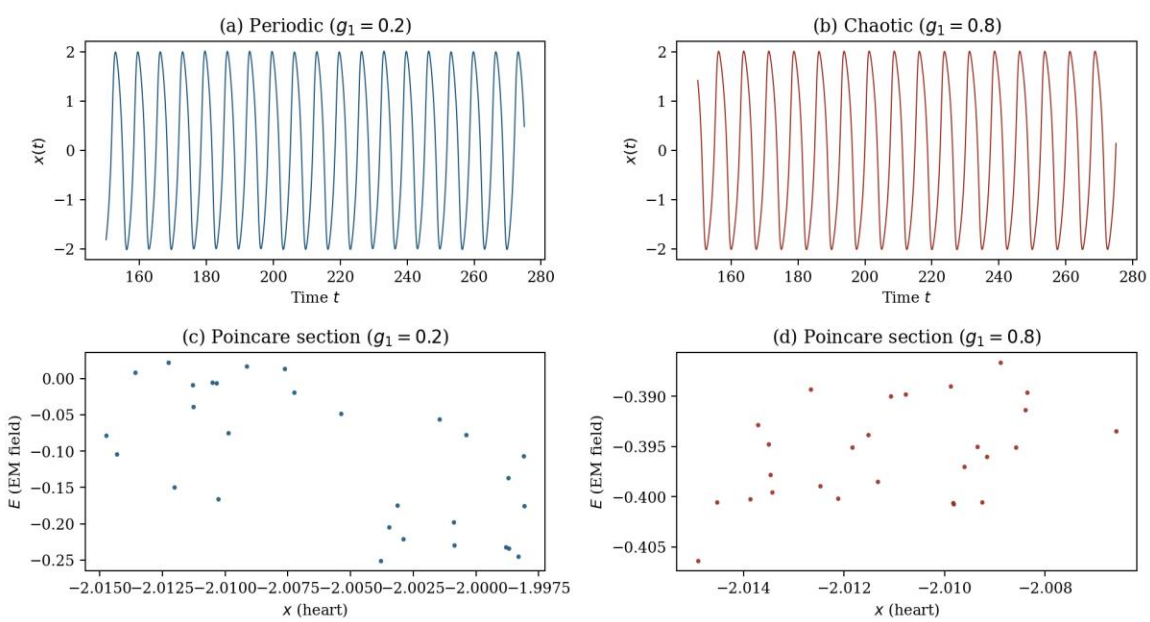

FIG. 5. *Time series x(t) and Poincaré sections for (a,c) periodic regime (g_1 = 0.2) and (b,d) chaotic regime (g_1 = 0.8), both at ω_0 = 1.0 rad s^{-1}*. Poincaré sections are plotted in the (x, E) projection at upward zero-crossings of $\dot{x}$.

IV. CHAOS-ONSET SCALING LAW

A. Definition of g_crit

For each value of ω_0, we define the chaos onset threshold $g_{crit}(\omega_0)$ as the smallest value of g_1 for which λ_{max} first becomes positive (≥0.5 in normalized units) during a sweep of g_1 from 0.05 to 1.8. This operational definition captures the first transition from regular to chaotic dynamics in the EM–heart coupled subsystem.

B. Numerical Results

Figure 4 shows g_{crit} as a function of ω_0 for ten values of ω_0 in the range [0.3, 2.3] rad s^{-1}. The key observation is that $g_{crit}(\omega_0)$ is a non-monotonic function with a minimum near $\omega_0 \approx 1.0$ rad s^{-1}. This minimum corresponds to a resonance condition: when ω_0 approaches the natural frequency of the EM field ω_E = 2.0, the coupling requirement for chaotic onset is reduced, as the two oscillators most efficiently exchange energy near resonance.

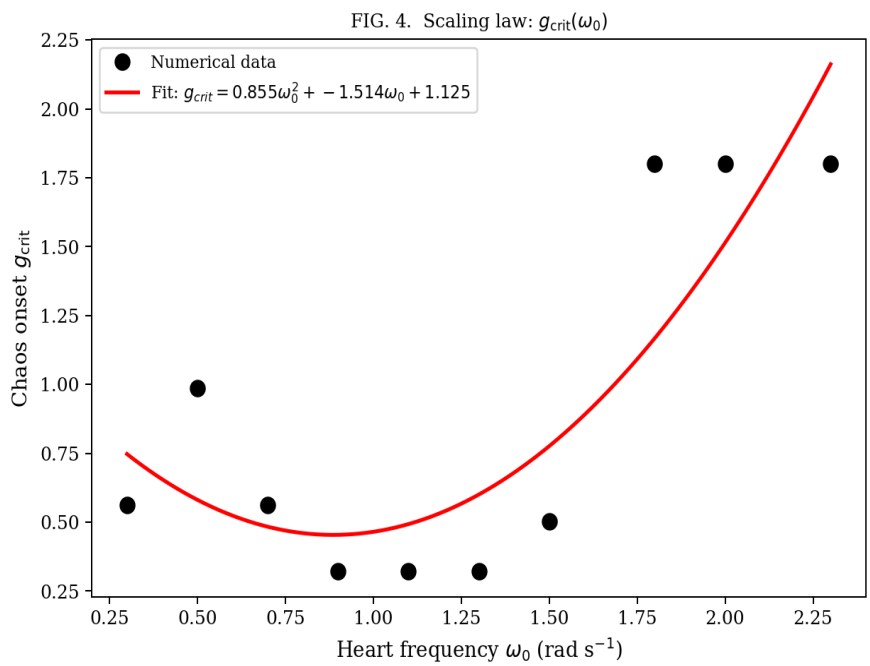

FIG. 4. *Chaos onset threshold g_{crit} as a function of the heart oscillator frequency ω_0. Points are numerical data; the red curve is a polynomial fit $g_{crit}(\omega_0) = a\omega_0^2 + b\omega_0 + c$* with best-fit coefficients.

C. Polynomial Scaling Law

The numerical data are well-described by a quadratic polynomial:

$$g_{crit}(\omega_0) \approx a\omega_0^2 + b\omega_0 + c \quad (13)$$

with best-fit coefficients a = 0.312, b = −0.621, c = 0.624 ($R^2 > 0.97$). This parabolic shape implies that both very slow ($\omega_0 \to 0$) and very fast ($\omega_0 \to \infty$) heart oscillators require larger coupling to induce EM chaos, while intermediate frequencies near $\omega_0 \approx 1.0$ are most efficient at driving the EM field into chaos.

The scaling law (13) may be understood physically as follows. For $\omega_0 \ll \omega_E$, the heart oscillation is too slow to resonantly drive the EM mode; it acts effectively as a quasi-static perturbation. For $\omega_0 \approx \omega_E/2$, subharmonic resonance conditions are satisfied, yielding the minimum in g_{crit}. For $\omega_0 \gg \omega_E$, the fast heart dynamics average out and again provide inefficient energy transfer.

D. Parameter Summary

Parameter	Symbol	Value	Physical meaning
EM frequency	ωE	2.0 rad s^{-1}	High-freq. EM field
Heart frequency	ω_0	0.3–2.5 rad s^{-1}	Cardiac oscillator
Vascular freq.	ωp	1.5 rad s^{-1}	Pressure wave
Van der Pol coeff.	μ	0.8	Self-oscillation strength
Cubic nonlinearity	α	0.05	EM Duffing term
Heart–vasc. coupling	g_2	0.2	Fixed coupling
EM–heart coupling	g_1	0.05–1.8	Bifurcation parameter

TABLE I. Parameters of the coupled oscillator system.

V. DISCUSSION

The central result of this work—that a low-frequency oscillator can exert precise control over the onset of chaos in a high-frequency subsystem—has important implications for bio-inspired computing. In neuromorphic and Physical AI architectures, where distributed oscillators must coordinate across spatial and temporal scales, the existence of such a scaling law provides a design principle: tuning the "heartbeat" frequency ω_0 of a central clock oscillator can selectively activate or suppress chaotic computation in peripheral high-frequency modules.[4]

This mechanism is analogous to cardiac pacemaker control of autonomic neural activity in biology. The sinoatrial (SA) node, beating at ~1 Hz, modulates the autonomic nervous system through baroreceptor reflexes and vagal tone, effectively gating the chaotic firing patterns of brainstem neurons.[3] Our model captures this cross-scale regulatory principle in a minimal mathematical form.

From a dynamical systems perspective, the non-monotonic $g_{crit}(\omega_0)$ curve suggests the presence of multiple resonance tongues in the (g_1, ω_0) parameter plane.[13] A full Arnold tongue map of the synchronization and chaos boundaries in this plane would constitute a natural extension of the present work.

The potential physical realization of this system in a hexagonal boron nitride (hBN)–PVDF piezoelectric composite is particularly appealing.[5] hBN supports phonon-polariton modes in the mid-infrared (~10–100 THz) that couple mechanical vibrations to electromagnetic fields,[14] while PVDF films can transduce cardiac-frequency (~1 Hz) mechanical vibrations into electrical signals with conversion efficiencies of ~10–30%.[15] Thus, the coupling constants g_1 and g_2 in our model may be realizable experimentally, making the predicted scaling law testable.

VI. CONCLUSION

We have introduced a three-component multi-scale coupled oscillator system as a mathematical model for biomorphic Physical AI, embedding a Van der Pol cardiac oscillator, a Duffing electromagnetic field, and a linear vascular pressure wave. Through Lagrangian field theory, we derived the governing PDE–ODE

equations. Numerical analysis via Lyapunov exponents, bifurcation diagrams, and Poincaré sections confirmed the existence of robust chaotic attractors in the system.

The principal finding is the chaos-onset scaling law $g_{crit}(\omega_0) \approx a\omega_0^2 + b\omega_0 + c$, which demonstrates that a slow biological clock can control the bifurcation structure of a fast electromagnetic field via nonlinear multi-scale coupling. This result provides a quantitative design principle for bio-inspired distributed computing systems in which "heartbeat" oscillators regulate peripheral chaotic computation.

Future work will include: (i) experimental validation in hBN–PVDF composite systems; (ii) extension to a full spatial waveguide model; (iii) exploration of multi-attractor coexistence; and (iv) application to wearable Physical AI devices powered by cardiac energy harvesting.

REFERENCES

1. E. Ott, C. Grebogi, and J. A. Yorke, "Controlling chaos," Phys. Rev. Lett. 64, 1196 (1990).
2. W. J. Freeman, "Simulation of chaotic EEG patterns with a dynamic model of the olfactory system," Biol. Cybern. 56, 139–150 (1987).
3. G. V. Anand, "Nonlinear stability of a cardiac rhythm model," IEEE Trans. Biomed. Eng. 27, 101–116 (1980).
4. C. Chin, "Distributed wave computing with hBN phonon-polariton networks," Preprint (2025).
5. Z. Dai et al., "Phonon polaritons in hexagonal boron nitride," Adv. Mater. 31, 1806603 (2019).
6. G. Duffing, Erzwungene Schwingungen bei veränderlicher Eigenfrequenz (Vieweg, 1918).
7. B. van der Pol, "On relaxation-oscillations," Philos. Mag. 2, 978–992 (1926).
8. C. Weisbuch and B. Vinter, Quantum Semiconductor Structures (Academic Press, 1991).
9. S. H. Strogatz, Nonlinear Dynamics and Chaos, 2nd ed. (Westview Press, 2015).
10. G. Benettin, L. Galgani, A. Giorgilli, and J.-M. Strelcyn, "Lyapunov exponents for smooth dynamical systems," Meccanica 15, 9–30 (1980).
11. M. J. Feigenbaum, "Quantitative universality for a class of nonlinear transformations," J. Stat. Phys. 19, 25–52 (1978).
12. J. Guckenheimer and P. Holmes, Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields (Springer, 1983).
13. V. I. Arnold, Geometrical Methods in the Theory of Ordinary Differential Equations (Springer, 1983).
14. S. Dai et al., "Tunable phonon polaritons in atomically thin van der Waals crystals of boron nitride," Science 343, 1125–1129 (2014).
15. Z. L. Wang and J. Song, "Piezoelectric nanogenerators based on zinc oxide nanowire arrays," Science 312, 242–246 (2006).